3DWEBSOFT INC. FOUNDATION

FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

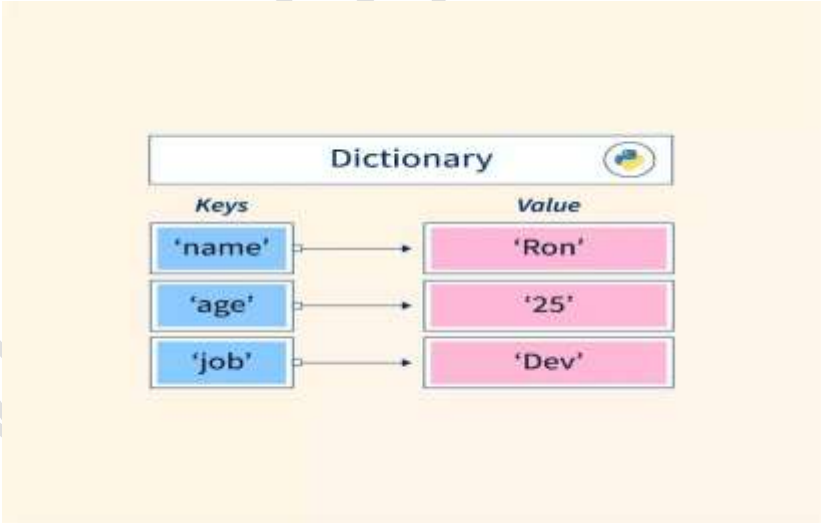
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 29: Dictionaries in Python



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What is a Dictionary?

A **Dictionary** is a collection of **key-value pairs**.

- Each **key** is unique.
- Each **key** has a **value**.
- Dictionaries are written using **curly braces { }**.

- Dictionaries are **mutable**, which means you can change, add, or remove items.

Example

```
student = {
   "name": "Rahul",
   "age": 13,
   "class": "8th"
}


print(student)
```

Output

```
{'name': 'Rahul', 'age': 13, 'class': '8th'}
```

Real-Life Example

Think of a **student ID card**.

Key	Value
Name	Rahul
Age	13
Class	8th
School	ABC High School

Each piece of information has a **label (key)** and a **value**.

Features of Dictionaries

✅ Store data as **key-value pairs**

✅ Keys must be unique

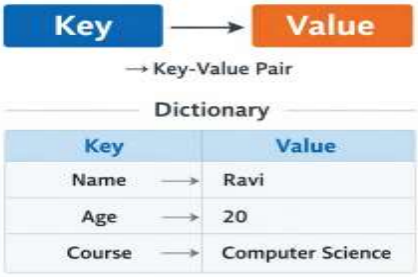
✅ Values can be repeated

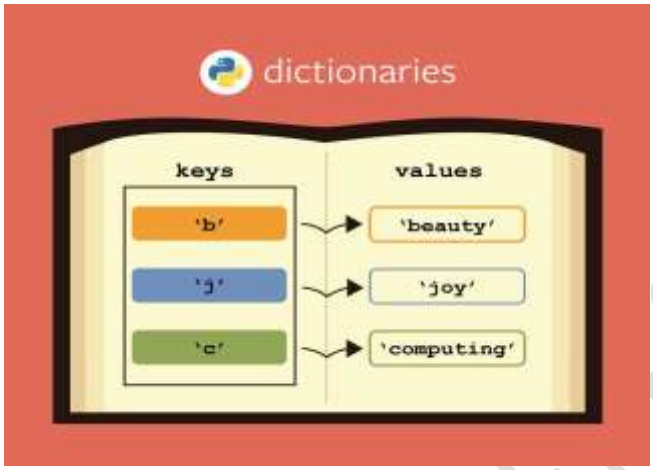
✅ Easy to search using keys

✅ Can store different data types

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Dictionary Structure







Dictionary

{

Name → Rahul

Age → 13

Class → 8th

}

Dictionary Syntax

```
dictionary_name = {
   "key1": value1,
```

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```
    "key2": value2
}
```

Example

```
car = {
    "brand": "Toyota",
    "color": "White"
}
```

Creating a Dictionary

```
student = {
    "name": "Anita",
    "age": 12,
    "class": "7th"
}

print(student)
```

Output

```
{'name': 'Anita', 'age': 12, 'class': '7th'}
```

Accessing Dictionary Values

```
student = {
    "name": "Rahul",
    "age": 13
}

print(student["name"])
```

Output

```
Rahul
```

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Using get() Method

```
student = {
   "name": "Rahul",
   "age": 13
}

print(student.get("age"))
```

Output

```
13
```

Adding a New Item

```
student = {
   "name": "Rahul",
   "age": 13
}

student["city"] = "Hyderabad"

print(student)
```

Output

```
{'name': 'Rahul', 'age': 13, 'city': 'Hyderabad'}
```

Changing a Value

```
student = {
   "name": "Rahul",
   "age": 13
}
```

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```
student["age"] = 14


print(student)
```

Output

```
{'name': 'Rahul', 'age': 14}
```

Removing an Item

```
student = {
    "name": "Rahul",
    "age": 13
}


student.pop("age")


print(student)
```

Output

```
{'name': 'Rahul'}
```

Finding Dictionary Length

```
student = {
    "name": "Rahul",
    "age": 13,
    "class": "8th"
}


print(len(student))
```

Output

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Printing Keys

```
student = {
   "name": "Rahul",
   "age": 13,
   "class": "8th"
}
print(student.keys())
```

Output

```
dict_keys(['name', 'age', 'class'])
```

Printing Values

```
student = {
   "name": "Rahul",
   "age": 13,
   "class": "8th"
}

print(student.values())
```

Output

```
dict_values(['Rahul', 13, '8th'])
```

Printing Keys and Values

```
student = {
   "name": "Rahul",
   "age": 13
}
```

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```
print(student.items())
```

Output

```
dict_items([('name', 'Rahul'), ('age', 13)])
```

Loop Through Keys

```
student = {
   "name": "Rahul",
   "age": 13
}

for key in student:
   print(key)
```

Output

```
name
age
```

Loop Through Values

```
student = {
   "name": "Rahul",
   "age": 13
}

for value in student.values():
   print(value)
```

Output

```
Rahul
13
```

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Loop Through Keys and Values

```
student = {
   "name": "Rahul",
   "age": 13
}

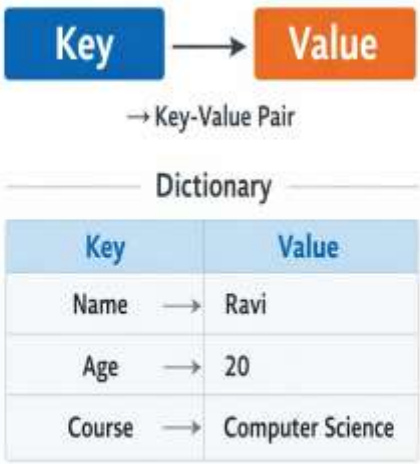
for key, value in student.items():
   print(key, ":", value)
```

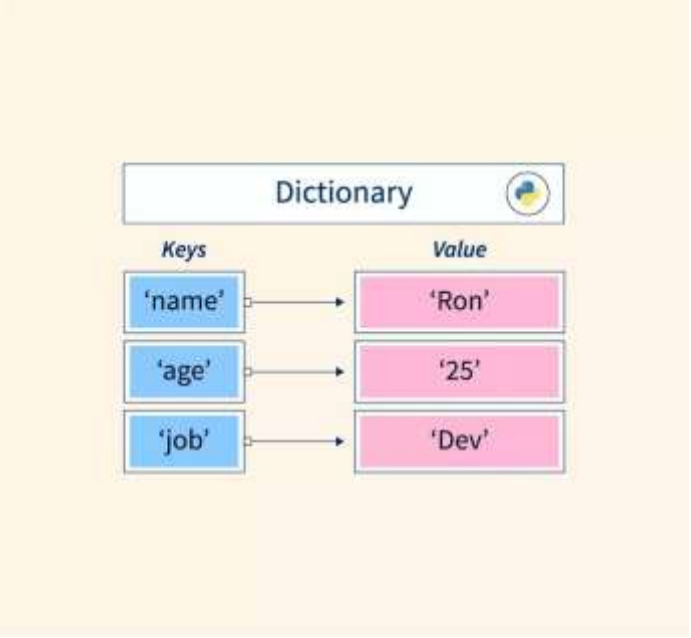
Output

name : Rahul

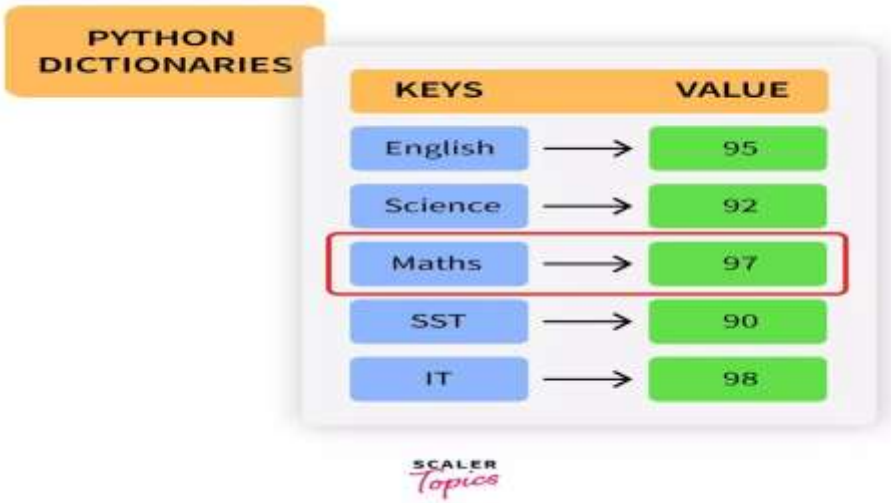
age : 13

Dictionary Key & Value





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Key Value

Name → Rahul

Age → 13

Class → 8th

Checking if a Key Exists

```
student = {
   "name": "Rahul",
   "age": 13
}

print("name" in student)
```

Output

True

Copying a Dictionary

```
student = {
   "name": "Rahul",
```

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```
    "age": 13
}

new_student = student.copy()

print(new_student)
```

Clearing a Dictionary

```
student = {
    "name": "Rahul",
    "age": 13
}

student.clear()

print(student)
```

Output

```
{}
```

Nested Dictionary

```
students = {
    "student1": {
        "name": "Rahul",
        "age": 13
    },
    "student2": {
        "name": "Anita",
        "age": 12
```

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```
    }
}

print(students)
```

Difference Between List, Tuple, and Dictionary

List	Tuple	Dictionary
Uses []	Uses ()	Uses {}
Ordered	Ordered	Key-Value Pairs
Mutable	Immutable	Mutable
Uses Index	Uses Index	Uses Keys

Real-Life Examples

Student Information

```
student = {
    "name": "Rahul",
    "class": "8th",
    "marks": 95
}
```

Book Details

```
book = {
    "title": "Python Basics",
    "author": "ABC",
    "price": 250
}
```

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Mobile Phone

```
phone = {
   "brand": "Samsung",
   "model": "A15",
   "color": "Black"
}
```

Practical Session

Practical Exercise 1

```
fruit = {
   "name": "Apple",
   "color": "Red"
}

print(fruit)
```

Practical Exercise 2

```
student = {
   "name": "Rahul",
   "age": 13
}

print(student["age"])
```

Practical Exercise 3

```
student = {
   "name": "Rahul"
```

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```
}

student["city"] = "Hyderabad"

print(student)
```

Practical Exercise 4

```
student = {
    "name": "Rahul",
    "age": 13
}

for key, value in student.items():
    print(key, value)
```

Practical Exercise 5

```
student = {
    "name": "Rahul",
    "age": 13
}

print(len(student))
```

Mini Activity

Guess the Output

```
car = {
    "brand": "Toyota",
```

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```
    "color": "White"
}

print(car["color"])
```

Answer

White

Guess the Output

```
student = {
    "name": "Rahul"
}

student["age"] = 13

print(student)
```

Answer

{'name': 'Rahul', 'age': 13}

Classroom Activity

Create a dictionary to store:

- Your Name
- Age
- Class
- School
- Favorite Subject

Print all keys and values using a **for loop**.

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Quiz

Choose the Correct Answer

1. Which brackets are used for dictionaries?

A) []

B) ()

C) {}

D) <>

✅ **Answer:** C

2. A dictionary stores data as

A) Numbers only

B) Key-Value Pairs

C) Letters only

D) Lists only

✅ **Answer:** B

3. Which method returns all keys?

A) values()

B) keys()

C) items()

D) get()

✅ **Answer:** B

4. Which method removes a key?

A) remove()

B) delete()

C) pop()

D) erase()

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✅ **Answer:** C

5. Which method returns both keys and values?

A) items()

B) values()

C) keys()

D) copy()

✅ **Answer:** A

Homework

1. Create a dictionary with your personal details.
2. Print your name using the key.
3. Add your favorite color.
4. Change your age.
5. Remove one item using pop().
6. Print all keys.
7. Print all values.
8. Print all key-value pairs.
9. Create a dictionary of 5 fruits and their colors.
10. Create a nested dictionary with details of two students.

Key Points to Remember

- A **Dictionary** stores information using **key-value pairs**.
- Dictionaries use **curly braces {}**.
- Keys

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